

UDL Beam

Example 1 - rivt Doc

Attn: User Example

proj. 0001



Table of Contents

0.1 Summary and Loads	1
0.1 - 2 Load Combinations	1
0.1 - 3 Loads and Geometry	1
0.1 - 4 Beam Response	3



0.1 | Summary and Loads

This rivt file example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam using E-B theory ¹. It also serves as an annotated example of a single rivt doc with multiple sections that is not part of a report.

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the < [rivt user manual](#) > .

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command at the end of each rivt file (Doc-API rv.D). Published files are found in the _published folder.

0.1 - 2 | Load Combinations

Dead and live loads effects are taken from ASCE 7-05 ²

Table 1: Load Effects (stored: t001-1.csv)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

0.1 - 3 | Loads and Geometry

Successive value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 2: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 k_ft	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-05 LL
b_1	10.00 inch	254.00 mm	beam width
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
Fb_1	20000.00 lb_in2	137.90 MPA	allowable stress

The VALTABLE command reads variable values from a file in the rvsr folder. The description is the table title, followed by the max column width.



Table 3: Beam Geometry (rvsrc/beam1.csv)

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing
spn_1	16.00 ft	4.88 m	beam span

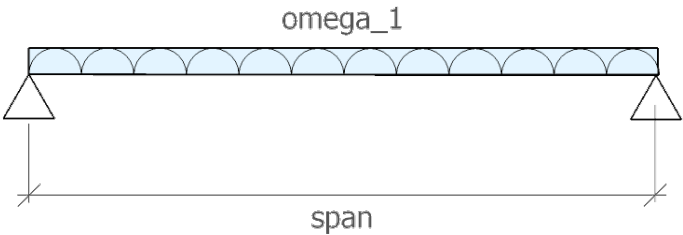


Fig. 1 - Beam Diagram

Uniform Distributed Loads

Eq. 1: Dead load [ASCE7-05 2.3.2]

$dl_1 = 1.2 \cdot (D_4 + spc_1 \cdot (D_1 + D_2 + D_3))$				
$dl_1 = 3.64 \text{ k_ft}$ $[dl_1] = 53.09 \text{ kN_m}$ Dead load [ASCE7-05 2.3.2]				
D_4	spc_1	D_1	D_2	D_3
3.00 k_ft	2.00 ft	3.80 psf	2.10 psf	10.00 psf
fixed machinery DL	beam spacing	joists DL	plywood DL	partitions DL

Eq. 2: Live load [ASCE7-05 2.3.2]

$ll_1 = 1.6 \cdot L_1 \cdot spc_1$	
$ll_1 = 0.13 \text{ k_ft}$ $[ll_1] = 1.87 \text{ kN_m}$ Live load [ASCE7-05 2.3.2]	
L_1	spc_1
40.00 psf	2.00 ft
ASCE7-05 LL	beam spacing



Eq. 3: Total load [ASCE7-05 2.3.2]

```

w1 = dl1 + ll1

w1 = 3.77 k_ft      [w1] = 54.96 kN_m    | Total load [ASCE7-05 2.3.2]

ll1                  dl1
-----
128.00 ft·psf        3.64 k_ft
-----
Live load [ASCE7-05 2.3.2]  Dead load [ASCE7-05 2.3.2]
-----

```

0.1 - 4 | Beam Response

The following lines import the beam geometry from an external file, calculate section properties from imported functions and calculate the maximum moment, bending stress and mid-span deflection.

Table 4: Beam functions (rvsrc/scripts/sectprop.py)

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

Eq. 4: rectangle - S (sectprop.py)

```

section1 = rectsect(b1, h1)

section1 = 540.00 in3      [section1] = 8849.01 cm3    | rectangle - S (sectprop.py)

h1          b1
-----
18.00 inch   10.00 inch
-----
beam depth   beam width
-----

```

Eq. 5: rectangle - I (sectprop.py)

```

inertia1 = rectinertia(b1, h1)

inertia1 = 4860.0 in4      [inertia1] = 202288.5 cm4    | rectangle - I (sectprop.py)

h1          b1
-----

```



<u>18.0 inch</u>	<u>10.0 inch</u>
<u>beam depth</u>	<u>beam width</u>

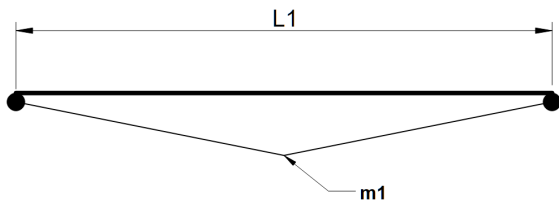


Fig. 2 - Moment diagram

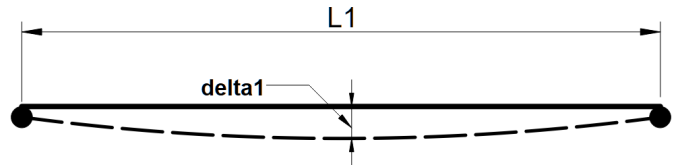


Fig. 3 - Deflection diagram

Maximum bending stress formula

Eq.6:

$$\sigma_1 = \frac{M_1}{S_1}$$

Eq. 7: Mid-span UDL moment

$$m_1 = \frac{\omega_1 \cdot \text{spn}_1^2}{8}$$

$m_1 = 120.52 \text{ ftkip}$ $[m_1] = 163.40 \text{ mkN}$ | Mid-span UDL moment

<u>spn₁</u>	<u>ω₁</u>
<u>16.00 ft</u>	<u>3.77 k_ft</u>
<u>beam span</u>	<u>Total load [ASCE7-05 2.3.2]</u>

Eq. 8: Bending stress



$$fb_1 = \frac{m_1}{section_1}$$

$fb_1 = 2678.2 \text{ lb_in}^2$ $[fb_1] = 18.5 \text{ MPA}$ | Bending stress

m_1	$section_1$
<u>120.5 ft²·k_ft</u>	<u>540.0 inch³</u>
Mid-span UDL moment	rectangle - S (sectprop.py)

Eq.9: Stress ratio

$fb_1 < Fb_1$				
[1] fb_1	[2] Fb_1	ratio [1]/[2]	check	reference
<u>2.68 ksi</u>	<u>20.00 ksi</u>	<u>0.13</u>	<u>OK</u>	<u>Stress ratio</u>

Eq. 10: mid-span deflection (sectprop.py)

$$\delta_1 = \text{midspan_}\delta(\text{spn}_1, w_1, E_1, \text{inertia}_1)$$

$\delta_1 = 0.04 \text{ inch}$ $[\delta_1] = 1.00 \text{ mm}$ | mid-span deflection (sectprop.py)

$inertia_1$	spn_1	w_1	E_1
<u>4860.00 inch⁴</u>	<u>16.00 ft</u>	<u>3.77 k_ft</u>	<u>29000.00 ksi</u>
rectangle - I (sectprop.py)	beam span	Total load [ASCE7-05 2.3.2]	modulus of elasticity

- 1 "Euler-Bernoulli beam theory," Wikipedia, Wikimedia Foundation. [Online].https://en.wikipedia.org/wiki/Euler-Bernoulli_beam_theory. [Accessed: Jun. 15, 2026].
- 2 ASCE/SEI 7-05, Minimum Design Loads for Buildings and Other Structures, American Society of Civil Engineers, 2005.

